MATH-101 Calculus and Analytical Geometry – 3-0 CHs

By

Prof Dr Safia Akram

LIMIT

Lec # 6

OBJECTIVE

- The concept of a “limit” is the fundamental building block on which all calculus concepts are based. In this lecture we will study limits informally, with the goal of developing an intuitive feel for the basic ideas.

- We will also develop theorems and procedures for calculating limits.

LIMITS AND CONTINUITY

OVERVIEW

The concept of a **limit** is a central idea that distinguishes calculus from algebra and trigonometry.

It is fundamental to **finding the tangent to a curve or the velocity of an object.**

LIMITS AND CONTINUITY

Definition of "LIMIT":

We write $lim_{x \to a} f(x) = L$ and say,

" the limit of $f(x)$ as x approaches a, $(\rightarrow)$ equals to "L".

Roughly speaking, this says that the values of f(x) becomes closer and closer to the number 'L' as x approaches to the number 'a' (from either side of a) but note equal to 'a' (i.e. $x \neq a$).

An alternative notation for $lim_{x \to a} f(x) = L$ is

$$f(x) \to L \text{ as } x \to a$$

which is usually read $f(x)$ approaches 'L' as "x" approaches 'a'.

DIRECT SUBSTITUTION PROPERTY

Definition of "LIMIT":

If f is a polynomial or a rational function and 'a' is in the domain of 'f' then $lim_{x \to a} f(x) = f(a)$.

Note:

Functions with the direct substitution property are called continuous at 'a'. However not all limits can be evaluated by direct substitution.

CALCULATING LIMITS USING "THE LIMIT LAWS"

This section presents theorems for calculating limits.

LIMITS LAWS:

1. SUM Rule.
2. Difference Rule.
3. Product Rule.
4. Constant Multiple Rule.
5. Quotient Rule.
6. Power Rule.
7. Constant Function.
8. Identity Function.

Now

Suppose that c is a constant and the limit $lim_{x \to a}\ f(x)$ and $lim_{x \to a}\ g(x)$ exist. Then

THE LIMIT LAWS

1. SUM Rule:

The limit of the sum of two functions is the sum of their limits.

$$lim_{x\to a}\,[f(x)+g(x)] = lim_{x\to a}\ f(x) + lim_{x\to a}\,g(x)$$

2. DIFFERENCE Rule:

The limit of the difference of two functions is the difference of their limits.

$$lim_{x\to a}\,[f(x)-g(x)] = lim_{x\to a}\ f(x) - lim_{x\to a}\,g(x)$$

3. PRODUCT Rule:

The limit of a Product of two functions is the product of their limits.

$$lim_{x\to a}\,[f(x)\ g(x)] = \left(lim_{x\to a}\ f(x)\right)\left(lim_{x\to a}\,g(x)\right)$$

4. **Constant Multiple Rule:**

$$lim_{x\to a}\,[cf(x)] = c\ lim_{x\to a}\ f(x)$$

5. **Quotient Rule:**

$$lim_{x\to a}\ \frac{f(x)}{g(x)} = \frac{lim_{x\to a}\ f(x)}{lim_{x\to a}\ \mathrm{g}(x)}\ if\ lim_{x\to a}\ g(x) \neq 0$$

6. **Power Rule:**

$$lim_{x\to a}\ (\ f(x))^n = \ (lim_{x\to a}\ f(x))^n$$

where 'n' is a +ve integer.

7. **Constant Function:**

If $f(x)$ is the constant function $\boldsymbol{f(x) = c}$, then $\boldsymbol{lim_{x\to a}\ c = c}$.

where 'c' is any constant number.

8. **Identity Function:**

If $f(x)$ is the identity function $\boldsymbol{f(x) = x}$, then for any value of 'a' $lim_{x\to a}\ x = a.$

Remark

1. $lim_{x \to a}\ x^n = a^n,$ where ‘n’ is a positive integer.

2. $lim_{x \to a} \sqrt[n]{x} = \sqrt[n]{a},$ where ‘n’ is a positive integer.

3. $lim_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{lim_{x \to a} f(x)},$

where ‘n’ is a positive integer

Examples

Example # 1 Find $\lim_{x \to c} (x^3 + 4x^2 - 3)$

Solution

$$\lim_{x \to c} (x^3 + 4x^2 - 3) = \lim_{x \to c} x^3 + \lim_{x \to c} 4x^2 - \lim_{x \to c} 3$$

(Using Sum and Difference Rule)

$$= c^3 + 4c^2 - 3.$$

Example # 2 Find $\lim_{x \to c} \dfrac{x^4 + x^2 - 1}{x^2 + 5}$

Solution

$$\lim_{x \to c} \frac{x^4 + x^2 - 1}{x^2 + 5} = \frac{\lim_{x \to c}(x^4 + x^2 - 1)}{\lim_{x \to c}(x^2 + 5)}$$

Quotient Rule

$$= \frac{\lim_{x \to c}(x^4) + \lim_{x \to c}(x^2) - \lim_{x \to c}(1)}{\lim_{x \to c}(x^2) + \lim_{x \to c}(5)}$$

(Sum & Difference Rule)

$$= \frac{(\lim_{x \to c} x)^4 + (\lim_{x \to c} x)^2 - (\lim_{x \to c} 1)}{(\lim_{x \to c} x)^2 + (\lim_{x \to c} 5)}$$

Power Rule

$$= \frac{c^4 + c^2 - 1}{c^2 + 5}$$

Example # 3 Find $lim_{x\to -2}\ \sqrt{4x^2 - 3}$

Solution $lim_{x\to -2}\ \sqrt{4x^2 - 3} = \sqrt{lim_{x\to -2}\,(4x^2 - 3)}$

$$= \sqrt{lim_{x\to -2}\,(4x^2) - lim_{x\to -2}\,(3)}$$

$$= \sqrt{4\,(lim_{x\to -2}\,x)^2 - lim_{x\to -2}\ (3)} = \sqrt{4(-2)^2 - 3}$$

$$= \sqrt{4(4) - 3} \qquad = \sqrt{16 - 3} = \sqrt{13}$$

THEOREM

1. Limits of Polynomials can be found by Substitution

If $$P(x) = a_n x^n + a_{n_1} x^{n-1} + \cdots + a_0$$

$$lim_{x \to c}\, P(x) = P(c) = a_n c^n + a_{n_1} c^{n-1} + \cdots + a_0$$

2. Limits of Rational functions can be found by substitution if the limit of the Denominator is not zero.

If $P(x)$ and $Q(x)$ are polynomials and $Q(c) \neq 0$ then

$$lim_{x \to c}\, \frac{P(x)}{Q(x)} = \frac{P(c)}{Q(c)}$$

Examples (Limit of Rational Function)

Example # 1 Find $lim_{x\to -1}\ \frac{x^3+4x^2-3}{x^2+5}$

Solution

$$lim_{x\to -1}\ \frac{x^3+4x^2-3}{x^2+5} = \frac{lim_{x\to -1}(x^3+4x^2-3)}{lim_{x\to -1}(x^2+5)}$$

$$= \frac{(-1)^3+4(-1)^2-3}{(-1)^2+5} = \frac{0}{6} = 0$$

Example # 2 Find $lim_{x\to 2}\ \frac{2x^2+1}{x^2+6x-4}$

Solution

$$lim_{x\to 2}\ \frac{2x^2+1}{x^2+6x-4} = \frac{lim_{x\to 2}(2x^2+1)}{lim_{x\to 2}(x^2+6x-4)}$$

$$= \frac{2(2)^2+1}{(2)^2+6(2)-4} = \frac{9}{12} = \frac{3}{4}$$

Evaluate the following Limits.

Question # 1 $\lim_{x\to -2} \frac{x^3+2x^2-1}{5-3x}$

$$\lim_{x\to -2} \frac{x^3+2x^2-1}{5-3x} = \frac{\lim_{x\to -2}(x^3+2x^2-1)}{\lim_{x\to -2}(5-3x)}$$

$$= \frac{\lim_{x\to -2}(x^3)+\lim_{x\to -2}(2x^2)-\lim_{x\to -2}(-1)}{\lim_{x\to -2}(5)-\lim_{x\to -2}(3x)}$$

$$= \frac{(-2)^3+2(-2)^2-1}{5-3(-2)} = \frac{-1}{11}$$

Eliminating Zero (Denominators Algebraically)

- If the denominator of the rational function is zero, canceling common factors in the numerator and denominator may reduce the fraction to one whose denominator is no longer zero at 'c'.

- If this happens, we can find the limit by substitution in the simplified fraction.

Question # 1 Evaluate $lim_{x\to 1}\ \frac{x^2+x-2}{x^2-x}$

Solution $lim_{x\to 1}\ \frac{x^2+x-2}{x^2-x}$

We can't substitute $x = 1$, because it makes the denominator zero

$$lim_{x\to 1}\ \frac{(x-1)(x+2)}{x\,(x-1)} = lim_{x\to 1}\ \frac{(x+2)}{x}$$

$$= \frac{lim_{x\to 1}\ (x+2)}{lim_{x\to 1}\ x} = \frac{3}{1} = 3$$

Question # 2 Evaluate $lim_{x\to 2}\ \frac{x^2+x-6}{x-2}$

Solution $lim_{x\to 2}\ \frac{x^2+x-6}{x-2}$

$$lim_{x\to 2}\frac{x^2+x-6}{x-2}=\frac{lim_{x\to 2}(x^2+x-6)}{lim_{x\to 2}\,(x-2)}$$

$$=\frac{(2)^2+2-6}{2-2}$$ (Denominator vanishes)

Now consider,

$$\frac{lim_{x\to 2}(x^2+x-6)}{lim_{x\to 2}\,(x-2)}=\frac{lim_{x\to 2}\,(x+3)(x-2)}{lim_{x\to 2}\,(x-2)}$$

$$=lim_{x\to 2}\,(x+3)=lim_{x\to 2}\,(2+3)=5$$

Examples

Question # 1 Find $lim_{x\to 1}\frac{x^2-1}{x-1}$ Try Yourself Ans = 2

Question # 2 Find $lim_{x\to 3}\frac{x^2-6x+9}{x-3}$ Try Yourself Ans = 0

Question # 3 Find $lim_{x\to -4}\frac{2x+8}{x^2+4x-12}$ Try Yourself Ans = 0

Question # 4 Find $lim_{x\to 5}\frac{x^2-3x-10}{x^2-10x+25}$ Try Yourself

Answer = doesn't exist (after simplification)

Question # 5 Find $lim_{h\to 0}\frac{(3+h)^2-9}{h}$ Try Yourself Ans = 6

Creating & Canceling Common Factor / Limits Involving Radicals

Example: Find $\lim_{h\to 0} \frac{\sqrt{2+h}-\sqrt{2}}{h}$

Solution We cannot find the limit by substituting $h = 0$ and the numerator and denominator don't have obvious factors. However, we can create a common factor in the numerator by multiplying it (and the denominator) by the so-called conjugate expression. $\sqrt{2+h}-\sqrt{2}$ obtained by changing the sign b/w the square roots.

Now

$$\frac{\sqrt{2+h}-\sqrt{2}}{h} = \left(\frac{\sqrt{2+h}-\sqrt{2}}{h}\right)\left(\frac{\sqrt{2+h}+\sqrt{2}}{\sqrt{2+h}+\sqrt{2}}\right) = \frac{2+h-2}{h\,(\sqrt{2+h}+\sqrt{2})}$$

$$= \frac{h}{h\,(\sqrt{2+h}+\sqrt{2})} = \frac{1}{\sqrt{2+h}+\sqrt{2}}$$

$$\lim_{h\to 0} \frac{\sqrt{2+h}-\sqrt{2}}{h} = \lim_{h\to 0} \frac{1}{\sqrt{2+h}+\sqrt{2}} = \frac{1}{\sqrt{2}+\sqrt{2}} = \frac{1}{2\sqrt{2}}$$

Examples

Question # 1: Find $lim_{x\to 1}\frac{x-1}{\sqrt{x}-1}$ Try Yourself Ans = 2

Question # 2: Find $\lim_{t\to 0}\frac{\sqrt{t^2+9}-3}{t^2}$ Try Yourself Ans = $\frac{1}{6}$

Question # 3: Find $lim_{x\to 0}\frac{\sqrt{x^2+100}-10}{x^2}$ Try Yourself Ans = $\frac{1}{20}$

Left-Hand Limit & Right-Hand Limit
or
Limits of a Piece-wise Defined Functions

For functions that are defined piece-wise, a two-sided limit at a point where the formula changes is best obtained by first finding the one-sided limits at that point.

Theorem

A function $f(x)$ has a limit as x approaches 'c' if and only if it has left hand and right-hand limits there and these one-sided limits are equal:

$$\lim_{x \to a} f(x) = L$$

$$\Leftrightarrow$$

$$\underbrace{\lim_{x \to a^+} f(x) = L}_{\text{RIGHT HAND LIMIT}} \quad \text{and} \quad \underbrace{\lim_{x \to a^-} f(x) = L}_{\text{LEFT HAND LIMIT}}$$

Example:

Evaluate $lim_{x\to 0} |x| = 0$

Solution Since according to the definition of $|x|$ we have

$$f(x) = |x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

$$lim_{x\to 0^-} f(x) = lim_{x\to 0^-} |x| = lim_{x\to 0^-} (-x) = 0$$

$$lim_{x\to 0^+} f(x) = lim_{x\to 0^+} |x| = lim_{x\to 0^+} (x) = 0$$

$$\Rightarrow L.H.L = R.H.L$$

so

$$lim_{x\to 0} |x| = 0 \ \ exist.$$

Example: Proof that $lim_{x\to 0} \frac{|x|}{x} = 0$ doesn't exist

Solution: Since according to the definition of $|x|$ we have

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

$$lim_{x\to 0^-} \frac{(-x)}{x} = lim_{x\to 0^-} (-1) = -1$$

$$lim_{x\to 0^+} \frac{(x)}{x} = lim_{x\to 0^+} (1) = 1$$

$$\Rightarrow L.H.L \neq R.H.L$$

So $lim_{x\to 0} \frac{|x|}{x} = 0$ doesn't exist

Examples

Question #1 Evaluate left- and right-hand limit of the function defined by

$$f(x) = \begin{cases} \dfrac{|x-3|}{x-3} & x \neq 3 \\ 0 & x = 3 \end{cases} \quad \text{at } x = 3$$

Solution: Try yourself

Question # 2 Let

$$f(x) = \begin{cases} \dfrac{1}{x+2} & x < -2 \\ x^2 - 5 & -2 < x \leq 3 \\ \sqrt{x+13} & x > 3 \end{cases}$$

Find

a) $\lim_{x \to -2} f(x)$

b) $\lim_{x \to 0} f(x)$

c) $\lim_{x \to 3} f(x)$

Solution: 2(a)

We will determine the stated two-sided limit by first considering the corresponding one-sided limits. For each one-sided limit, we must use that part of the formula that is applicable on the interval over with x −varies.

For example, as x approaches -2 from the left, the applicable part of the formula is

$$f(x) = \frac{1}{x+2}$$

and as x approaches -2 from the right, the applicable part of the formula near -2 is

$$f(x) = x^2 - 5$$

Thus $lim_{x\to -2^-} f(x) = lim_{x\to -2^-} \left(\frac{1}{x+2}\right) = -\infty$

$lim_{x\to -2^+} f(x) = lim_{x\to -2^+} (x^2 - 5) = -1$

Thus it follows that $lim_{x\to -2} f(x)$ doesn't exist.

Solution: 2(b)

The applicable part of the formula is $f(x) = x^2 - 5$ on both sides of 0, so there is no need to consider one-sided limits here.

$$lim_{x\to\ 0}\ f(x) = lim_{x\to 0}\ (x^2 - 5) = 0 - 5 = -5$$

Solution: 2(c)

Using the applicable parts of the formula for $f(x)$ We obtain

$$lim_{x\to 3^-} f(x) = lim_{x\to 3^-}\ (x^2 - 5) = 9 - 5 = 4$$

$$lim_{x\to 3^+} f(x) = lim_{x\to 3^+}\ \left(\sqrt{x + 13}\right) = \sqrt{3 + 13} = 4$$

Since the one-sided limits are equal so we have

$$lim_{x\to\ 3}\ f(x) = 4$$

Theorem

If $f(x)$ and $g(x)$ are two functions. If $f(x) \leq g(x)$ when x is near a (except possibly at a) and the limit of 'f' and 'g' both exist as x approaches 'a' then

$$lim_{x \to a}\, f(x) \leq lim_{x \to a}\, g(x)$$

SQUEEZE THEOREM
(Sandwich or Pinching Theorem)

If $f(x) \leq g(x) \leq h(x)$ when x is near a (except possibly at a) and

$$lim_{x \to a}\, f(x) = lim_{x \to a}\, h(x) = L$$

then

$$lim_{x \to a}\, g(x) = L$$

Example: Show that $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = 0$

Solution: First note that we cannot use

$\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = (\lim_{x\to 0} x^2)\,(\lim_{x\to 0} \sin\left(\frac{1}{x}\right))$ because $\lim_{x\to 0} \sin\left(\frac{1}{x}\right)$ doesn't exist. However

$$-1 \le \sin x \le 1$$

$$-1 \le \sin\left(\frac{1}{x}\right) \le 1$$

$$-x^2 \le x^2 \sin\left(\frac{1}{x}\right) \le x^2$$

We know that

$$\lim_{x\to 0} x^2 = 0 \quad \text{and} \quad \lim_{x\to 0} (-x^2) = 0$$

Taking $f(x) = -x^2$, $g(x) = x^2 \sin\left(\frac{1}{x}\right)$ and $h(x) = x^2$

So using SQUEEZE THEOREM

$$\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = 0$$

Example: Given that $1 - \frac{x^2}{4} \leq u(x) \leq 1 + \frac{x^2}{2}$ for all $x \neq 0$. Find $lim_{x \to 0}\ u(x)$, no matter how complicated u is.

Solution:

$$lim_{x \to 0} \left(1 - \frac{x^2}{4}\right) = 1 - 0 = 1$$

$$lim_{x \to 0} \left(1 + \frac{x^2}{2}\right) = 1 + 0 = 1$$

The sandwich theorem implies that

$$lim_{x \to 0}\ u(x) = 1$$

Exercises

Q#1

Given that $\lim_{x \to a} f(x) = -3$, $\lim_{x \to a} g(x) = 0$ & $\lim_{x \to a} h(x) = 8$. Find the limits that exist. If the limit doesnot exist, explain why.

a) $\lim_{x \to a} [f(x) + h(x)]$ $\xrightarrow{\text{Ans}}$ 5

b) $\lim_{x \to a} [f(x)]^2$ $\xrightarrow{\text{Ans.}}$ 9.

c) $\lim_{x \to a} \sqrt[3]{h(x)}$ $\xrightarrow{\text{Ans}}$ 2

d) $\lim_{x \to a} \frac{1}{f(x)}$ → Ans → $-1/3$

e) $\lim_{x \to a} \frac{f(x)}{h(x)}$ → Ans → $-3/8$

f) $\lim_{x \to a} \frac{g(x)}{f(x)}$ → Ans → 0

g) $\lim_{x \to a} \frac{f(x)}{g(x)}$ → Ans → does not exist

h) $\lim_{x \to a} \frac{2 f(x)}{h(x) - f(x)}$ → Ans → $-6/11$

Q#2 Suppose $\lim_{x \to c} f(x) = 5$ and $\lim_{x \to c} g(x) = -2$.

Find

a) $\lim_{x \to c} f(x)\, g(x)$

(b) $\lim_{x \to c} 2 f(x) g(x)$

c) $\lim_{x \to c} (f(x) + 3g(x))$

(d) $\lim_{x \to c} \dfrac{f(x)}{f(x) - g(x)}$

Q #3 Evaluating the limit and justify each step by indicating the appropriate Limit Law(s).

a) $\lim_{x \to 4} (5x^2 - 2x + 3)$

(b) $\lim_{x \to 2} \dfrac{2x^2 + 1}{x^2 + 6x - 4}$

c) $\lim_{t \to -2} (t+1)^9 (t^2 - 1)$

(d) $\lim_{x \to 1} \left(\dfrac{1 + 3x}{1 + 4x^2 + 3x^4} \right)^3$

e) $\lim_{t \to 6} 8(t-5)(t-7)$

(f) $\lim_{x \to 5} \dfrac{4}{x - 7}$

g) $\lim_{y \to 2} \dfrac{y + 2}{y^2 + 5y + 6}$

(h) $\lim_{x \to 2/3} 3s(2s - 1)$

Q#4 Evaluate the limit, if it exists

a) $\lim_{x \to -4} \dfrac{x^2+5x+4}{x^2+3x-4}$

(b) $\lim_{x \to 1} \dfrac{x^3-1}{x^2-1}$

c) $\lim_{x \to 2} \dfrac{x^2-x+6}{x-2}$

(d) $\lim_{h \to 0} \dfrac{\sqrt{1+h}-1}{h}$

e) $\lim_{x \to 7} \dfrac{\sqrt{x+2}-3}{x-7}$

(f) $\lim_{x \to -4} \dfrac{\frac{1}{4}+\frac{1}{x}}{4+x}$

g) $\lim_{x \to -1} \dfrac{\sqrt{x^2+8}-3}{x+1}$

(h) $\lim_{x \to 1} \dfrac{x-1}{\sqrt{x+3}-2}$

(i) $\lim_{x \to -2} \dfrac{-2x-4}{x^3+2x^2}$

(j) $\lim_{t \to -1} \dfrac{t^2+3t+2}{t^2-t-2}$

Q#5 Find the limits

1) $\lim_{x \to -3} \dfrac{2 - \sqrt{x^2 - 5}}{x + 3}$

(2) $\lim_{x \to 4} \dfrac{4 - x}{5 - \sqrt{x^2 + 9}}$

3) $\lim_{x \to 9} \dfrac{\sqrt{x} - 3}{x - 9}$

(4) $\lim_{x \to 4} \dfrac{4x - x^2}{2 - \sqrt{x}}$

5) $\lim_{t \to 0} \left[\dfrac{1}{t} - \dfrac{1}{t^2 + t} \right]$

(6) $\lim_{h \to 0} \dfrac{(3+h)^{-1} - 3^{-1}}{h}$

7) $\lim_{v \to 2} \dfrac{v^3 - 8}{v^4 - 16}$

(8) $\lim_{x \to 4} \dfrac{4x - x^2}{2 - \sqrt{x}}$

Q#6 Using the Sandwich Theorem

i) If $\sqrt{5-2x^2} \leq f(x) \leq \sqrt{5-x^2}$ for $-1 \leq x \leq 1$, find $\lim_{x \to 0} f(x)$ (Ans $= \sqrt{5}$)

ii) If $2 - x^2 \leq g(x) \leq 2\cos x$ for all x, find $\lim_{x \to 0} g(x)$.

iii) If $1 \leq f(x) \leq x^2 + 2x + 2$ for $\forall x$, find $\lim_{x \to -1} f(x)$.

iv) Using Squeeze theorem, to show that

$$\lim_{x \to 0} x^2 \cos(20\pi x) = 0.$$

v) If $3x \leq f(x) \leq x^3 + 2$ for $0 \leq x \leq 2$,

evaluate $\lim_{x \to 1} f(x)$.

Q#7 Find the limit, if it exists. If the limit doesnot exist, explain why.

1) $\lim_{x \to 2} \frac{|x-2|}{x-2}$

(2) $\lim_{x \to 0^-} \left(\frac{1}{x} - \frac{1}{|x|} \right)$

3) $\lim_{x \to 0^+} \left(\frac{1}{x} - \frac{1}{|x|} \right)$

Q#8 Let $g(x) = \begin{cases} -x & \text{if } x \leq -1 \\ 1-x^2 & \text{if } -1 < x < 1 \\ x-1 & \text{if } x > 1 \end{cases}$

Evaluate each of the following limits, if it exists.

i) $\lim_{x \to 1^+} g(x)$

(ii) $\lim_{x \to 1} g(x)$

(iii) $\lim_{x \to 0} g(x)$

iv) $\lim_{x \to -1^-} g(x)$

(v) $\lim_{x \to -1^+} g(x)$

(vi) $\lim_{x \to -1} g(x)$

Q#9 Let $F(x) = \dfrac{x^2-1}{|x-1|}$

a) Find (i) $\lim_{x \to 1^+} F(x)$ (ii) $\lim_{x \to 1^-} F(x)$

b) Does $\lim_{x \to 1} F(x)$ exist?

Q#10 Let $F(x) = (x+3)\dfrac{|x+2|}{(x+2)}$

Find

i) $\lim_{x \to -2^+} (x+3)\dfrac{|x+2|}{(x+2)}$ Ans = 1

ii) $\lim_{x \to -2^-} (x+3)\dfrac{|x+2|}{(x+2)}$ Ans = -1

Q#11) Evaluate Left and right hand limit of the ftn defined by

$$f(x) = \begin{cases} \dfrac{|x-3|}{x-3} & x \neq 3 \\ 0 & x = 3 \end{cases}$$

at $x = 3$.

Q#12 Let $f(x) = \begin{cases} x+2 & \text{if } x \le -1 \\ ax^2 & \text{if } x > -1 \end{cases}$

Find 'a' so that $\lim_{x \to -1} f(x)$ exists.

Q#13 If $f(x) = \begin{cases} 3 & \text{if } x \le -2 \\ -\frac{1}{2}x^2 & \text{if } -2 < x < 2 \\ 3 & \text{if } x \ge 2 \end{cases}$

Find (i) $\lim_{x \to -2^-} f(x)$, (ii) $\lim_{x \to -2^+} f(x)$

iii) $\lim_{x \to 2^-} f(x)$, (iv) $\lim_{x \to 2^+} f(x)$

Ans (i) 3, (ii) -2, (iii) -2, (iv) 3.

Q#14 Let $f(x) = \begin{cases} x^2+3 & \text{if } x \leq 1 \\ x+1 & \text{if } x > 1 \end{cases}$

Find (a) $\lim_{x \to 1^+} f(x)$ $\overset{\text{Ans}}{\Rightarrow} 2$

(b) $\lim_{x \to 1^-} f(x)$ $\xrightarrow{\text{Ans}} 4$